Puru Singh

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EDUCATION

Graphic Era University — Dehradun, UK
Bachelor of Technology, Computer Science, GPA: 8.73 — 2020 - 2024

WORK EXPERIENCE

Data Engineer at Genpact — Bengaluru, KA
Business Analyst Gen-AI — Oct 2024 – Current

- **Worked on the end-to-end migration** of Manufacturing Domain's entire ETL pipelines (100+) from Greenplum (PostgreSQL) to **SparkSQL and PySpark on Databricks**, improving job runtimes up to 80% via logical flow enhancements and Spark/Databricks native optimizations.
- **Owned the development of backend architecture** for ~110 Business Objects (BO) reports integrated with **Oracle ERP systems**, streamlining complex reporting for key business stakeholders.
- **Orchestrated robust workflows** using Apache Airflow and Databricks native features to ensure scalable, production-grade data processing.
- **Engineered and Integrated Gen-AI and RAG systems** using Python to automate insight generation and improve data discoverability within the data ecosystem alongside traditional ETL Pipelines for implementing Generative Data for the existing tables utilizing LLMs and Databricks ML libraries and functionalities.
- **Developed and maintained the backend for Sales and Operations Planning (S&OP) -** monthly cross-functional business process, ensuring the integrity of high-fidelity data for critical KPIs.
- **Collaborated across cross-functional teams** to streamline migration across Ingestion, Transformation, and Visualization layers.

Internship at Genpact — Hyderabad, TL
Data Analyst and Gen AI Intern — Feb 2024 – July 2024

- **Developed a Python automation script** using **LangChain** to generate summaries of 10-K financial reports, resulting in an **85% speedup** compared to existing manual solutions.
- **Built a Financial Health Dashboard** for Consumer Banking using **Power BI**, providing a comprehensive overview of income and spending with automated goal suggestions.
- **Performed data preprocessing and cleaning** to extract actionable insights for banking sector stakeholders.

Internship at Defense Research and Development Organization (DRDO) — New Delhi, DL
Android Development and Security Intern — Jul 2023 – Aug 2023

- **Developed a background Android application** using Kotlin and Jetpack Compose to capture & forward system notifications via SMS.
- **Conducted security analysis** using MobSF (static and dynamic), identifying critical flaws and reducing total testing time by 70%.
- **Implemented real-time logging** and system-level API integrations (SmsManager) for secure notification monitoring.

PROJECTS

Loocle – A Self-Hosted Photo Backup and Web App (Google Photos Inspired)

- **Engineered a cloud-independent photo management system** replicating the Google Photos experience on local hardware.
- **Built a Python Flask web app** with fast metadata-based search, caching, and multi-shot facial recognition features.
- **Automated wireless Android backups** using ADB over Wi-Fi for seamless syncing between devices.
- Implemented automatic **metadata tagging using Local LLMs using Ollama** to associate detected faces with person names, enhancing organization and quick retrieval.

ACHIEVEMENTS

Top 50 in Coursera's first Employment Hackathon — Online - Jan 2023

3rd Position in Intel's OneAPI Hackathon - Machine Learning, SYCL Programming — IIT Roorkee, UK - Mar 2023

ADDITIONAL

Technical Skills: Databricks, Python, PySpark, Spark SQL, Apache Spark, Apache Airflow, GIT (Github), AWS, C++, Microsoft Power BI, Excel, Generative AI (LangChain, RAGs, Ollama), Machine Learning, AI-Enabled Features in Databricks, Microsoft CoPilot
Domains & Concepts: System Design, Backend Development, ETL Pipelines, Data Engineering, CI/CD, Cloud Infrastructure, Facial Recognition, Computer Vision
Certifications & Training:
Databricks - Data Engineer - Professional, Data Engineer - Associate**,** Generative AI Engineer - Associate**,**
AWS - DEA-C01 Certified Data Engineer - Associate**,** CLF-C01 Certified Cloud Practitioner, **Microsoft -** AI-900 Azure AI Fundamentals
Languages & Interests: English (Fluent), Hindi (Fluent) | Fast Touch Typing (172 WPM, Top 7%), Reading, PC Gaming, Beatboxing